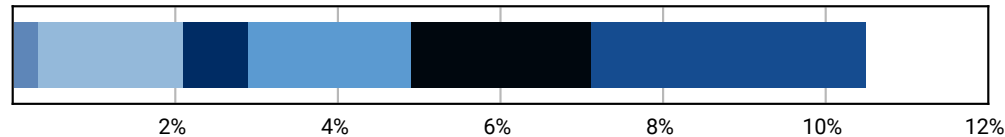


ADV: Statistics (Adv), S2 Interpretation and Bivariate Data (Adv) Bar Charts and Histograms (Y12)

Teacher: Denis Vlismas

Exam Equivalent Time: 67.5 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

S2 Interpretation and Bivariate Data



*Analytics based on the average contribution to the 2ADV/STD2 HSC exams over the past decade.

- Classifying Data
- Bar Charts and Histograms
- Other Chart Types
- Summary Statistics - Box Plots
- Summary Statistics - No Graph
- Bivariate Data Analysis

HISTORICAL CONTRIBUTION

- *S2 Interpretation and Bivariate Data* is a Year 12 topic that didn't previously exist in the Mathematics/Advanced course, although it has a decade long history in the Std2/Gen2 exam.
- It provides an area where examiners can test common content between the *Advanced* and *Standard 2* courses. The above bar chart is only included to show the relative importance of the subtopics in past Std2/Gen2 exams.
- *S2 Interpretation and Bivariate Data* has been split into six sub-topics for analysis which are listed in the bar chart above.
- This analysis looks at *Bar Charts and Histograms* (1.8%).

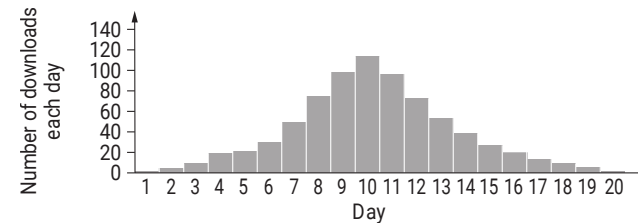
HSC ANALYSIS - What to expect and common pitfalls

- *Bar Charts & Histograms* were examined in the 2021 Advanced paper after being omitted in 2020.
- The more difficult *cumulative frequency histogram* questions are examined the most often, most recently in 2021 via a multiple choice question that was well answered. Past Std2 exams have however shown this topic can be quite challenging and we recommended a revision focus here.
- Note that the related *Pareto Charts* are likely to cannibalise some marks previously allocated to this sub-topic. These are covered under a separate sub-topic *Other Charts*.
- **Pitfalls:** Markers have consistently highlighted student difficulties in working with *class centres* and a very poor understanding of *finding the median and quartile values* of cumulative frequency histograms. With mean marks below 50% each of the last 4 times these topic areas have been examined, students can outperform peers by scoring highly here.

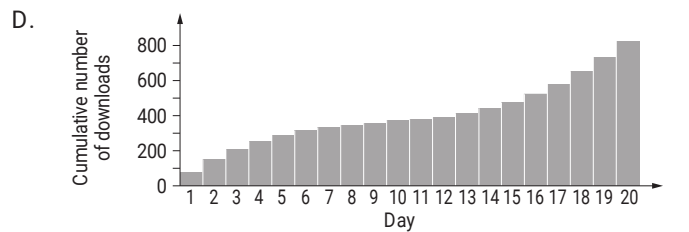
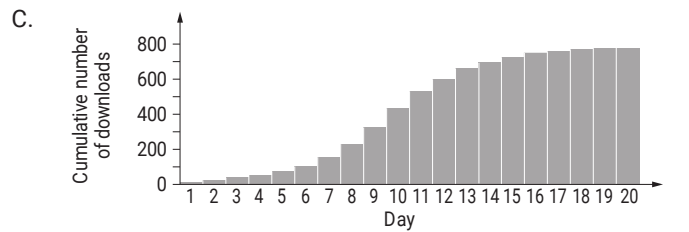
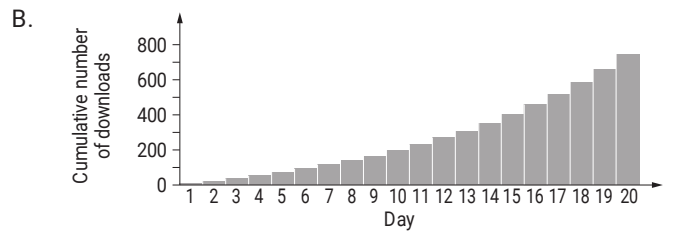
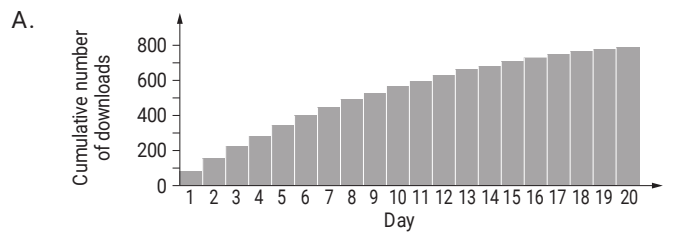
Questions

1. Statistics, 2ADV S2 2021 HSC 4 MC

The number of downloads of a song on each of twenty consecutive days is shown in the following graph.

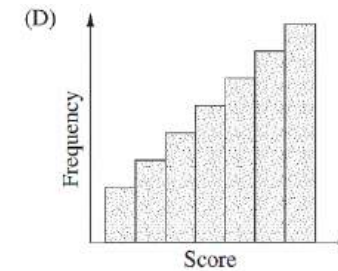
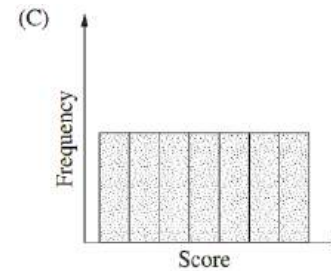
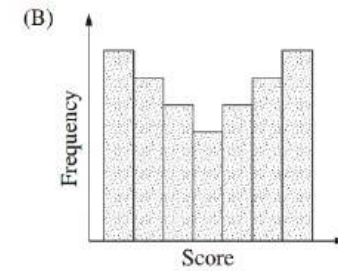
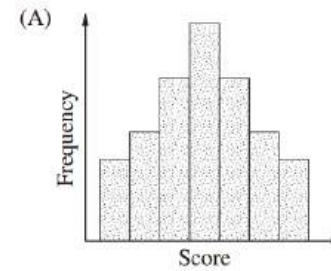


Which of the following graphs best shows the cumulative number of downloads up to and including each day?



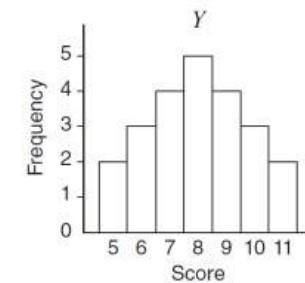
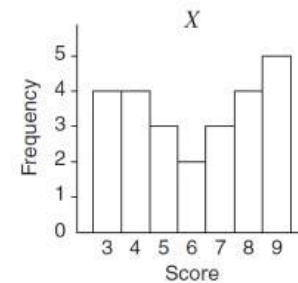
2. Statistics, STD2 S5 2010 HSC 4 MC

Which of the following frequency histograms shows data that could be normally distributed?



3. Statistics, STD2 S1 2011 HSC 11 MC

The sets of data, ***X*** and ***Y***, are displayed in the histograms.



Which of these statements is true?

- (A) ***X*** has a larger mode and ***Y*** has a larger range.
- (B) ***X*** has a larger mode and the ranges are the same.
- (C) The modes are the same and ***Y*** has a larger range.
- (D) The modes are the same and the ranges are the same.

4. Statistics, STD2 S1 2013 HSC 15 MC

The frequency histogram shows the number of goals scored by a football team in each game in a season.



What is the mean number of goals scored per game by this team?

- (A) 4
- (B) 4.5
- (C) 5
- (D) 5.5

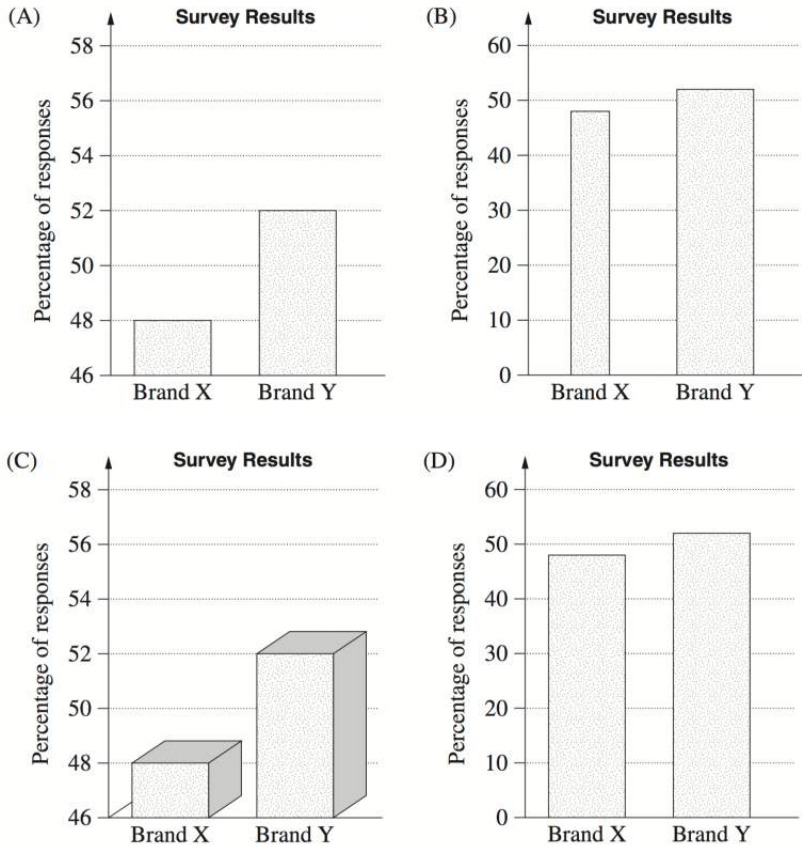
5. Statistics, STD2 S1 2013 HSC 6 MC

A survey was conducted where people were asked which of two brands of smartphones they preferred. The results were:

- 48% preferred Brand X
- 52% preferred Brand Y

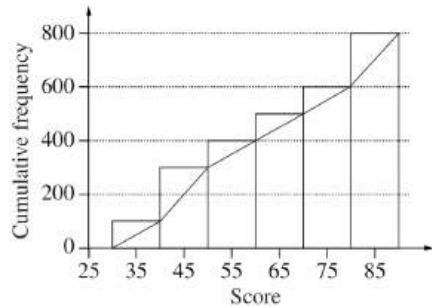
A graph displaying the data is to be included in a magazine article. The editor of the magazine wishes to ensure that the graph is not misleading in any way.

Which graph should the editor choose to include in the article?



6. Statistics, STD2 S1 2005 HSC 9 MC

A set of data is represented by the cumulative frequency histogram and ogive.

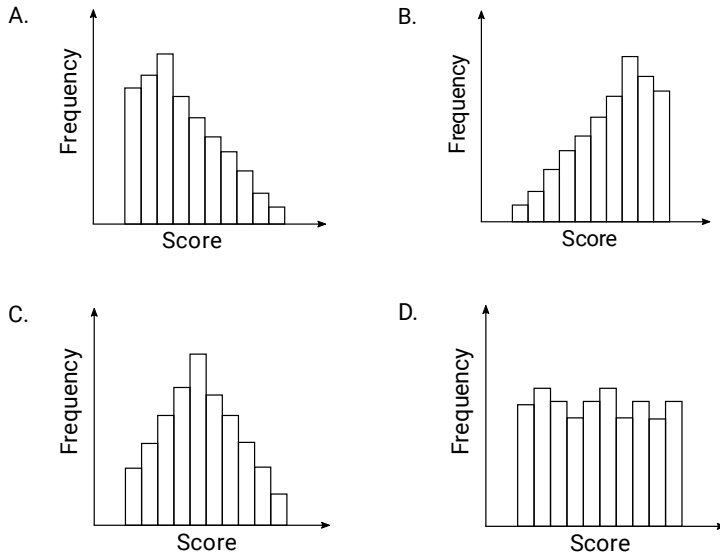


What is the best approximation for the interquartile range for this set of data?

- (A) 25
- (B) 30
- (C) 35
- (D) 40

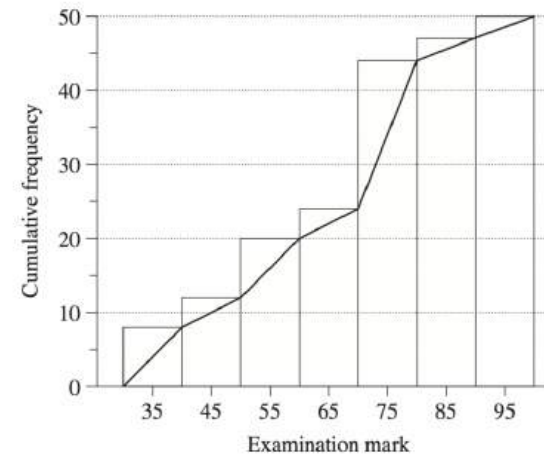
7. Statistics, STD2 S1 2020 HSC 7 MC

Which histogram best represents a dataset that is positively skewed?



8. Statistics, STD2 S1 2007 HSC 22 MC

A set of examination results is displayed in a cumulative frequency histogram and polygon (ogive).



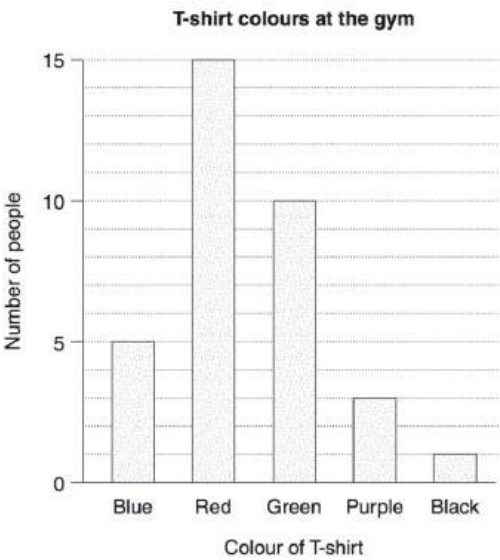
Sanath knows that his examination mark is in the 4th decile.

Which of the following could have been Sanath's examination mark?

- (A) 37
- (B) 57
- (C) 67
- (D) 77

9. Probability, STD2 S2 2010 HSC 23c

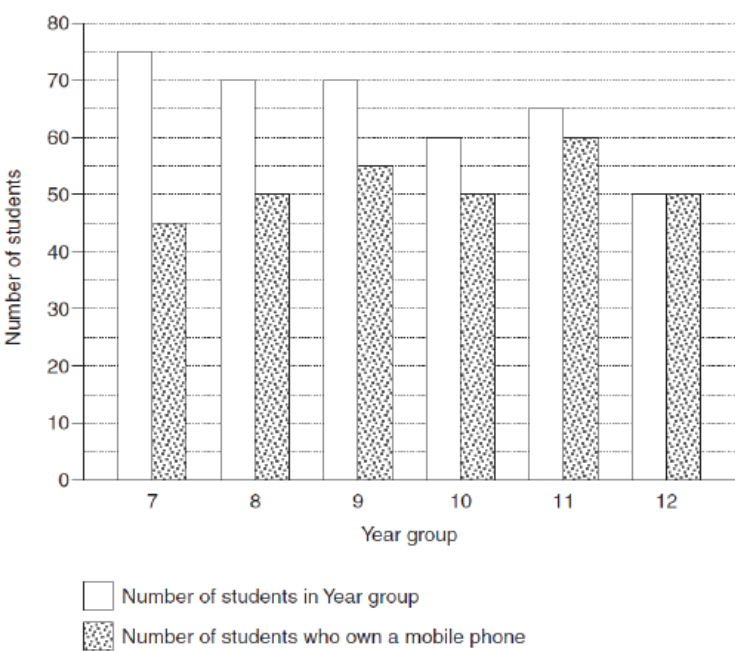
On Saturday, Jonty recorded the colour of T-shirts worn by the people at his gym. The results are shown in the graph.



- i. How many people were at the gym on Saturday? (Assume everyone was wearing a T-shirt). (1 mark)
- ii. What is the probability that a person selected at random at the gym on Saturday, would be wearing either a blue or green T-shirt? (1 mark)

10. Statistics, STD2 S1 2011 HSC 25b

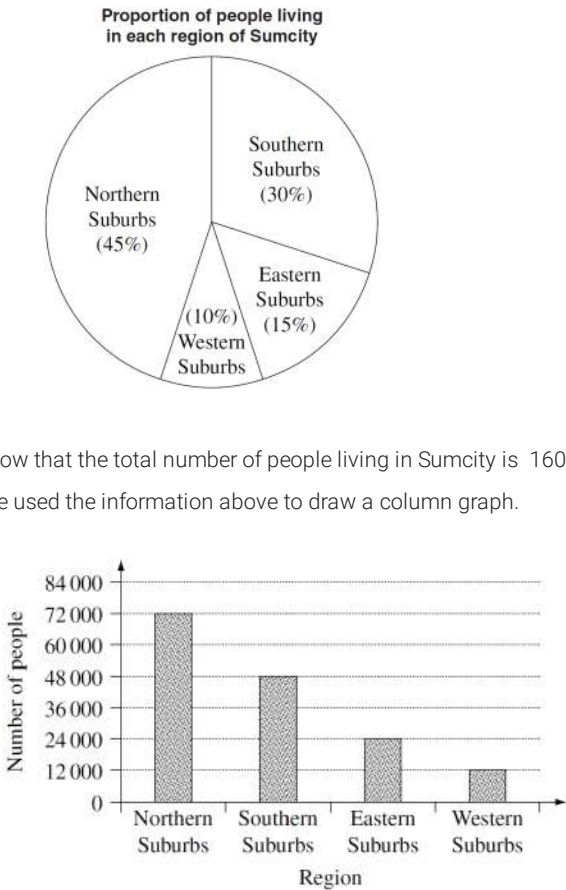
The graph below displays data collected at a school on the number of students in each Year group, who own a mobile phone.



- i. Which Year group has the highest percentage of students with mobile phones? (1 mark)
- ii. Two students are chosen at random, one from Year 9 and one from Year 10. Which student is more likely to own a mobile phone? Justify your answer with suitable calculations. (2 marks)
- iii. Identify a trend in the data shown in the graph. (1 mark)

11. Statistics, STD2 S1 2005 HSC 24d

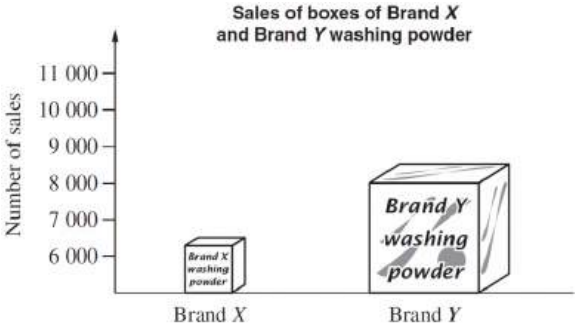
The sector graph shows the proportion of people, as a percentage, living in each region of Sumcity. There are 24 000 people living in the Eastern Suburbs.



- i. Show that the total number of people living in Sumcity is 160 000. (1 mark)
- Jake used the information above to draw a column graph.

- ii. The column graph height is incorrect for one region. Identify this region and justify your answer. (2 marks)

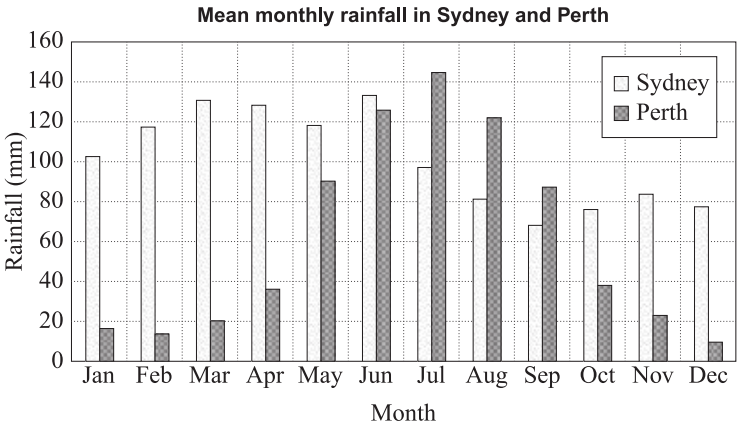
12. Statistics, STD2 S1 2006 HSC 24a



List TWO ways in which this graph is misleading. (2 marks)

13. Statistics, STD2 S1 2018 HSC 26d

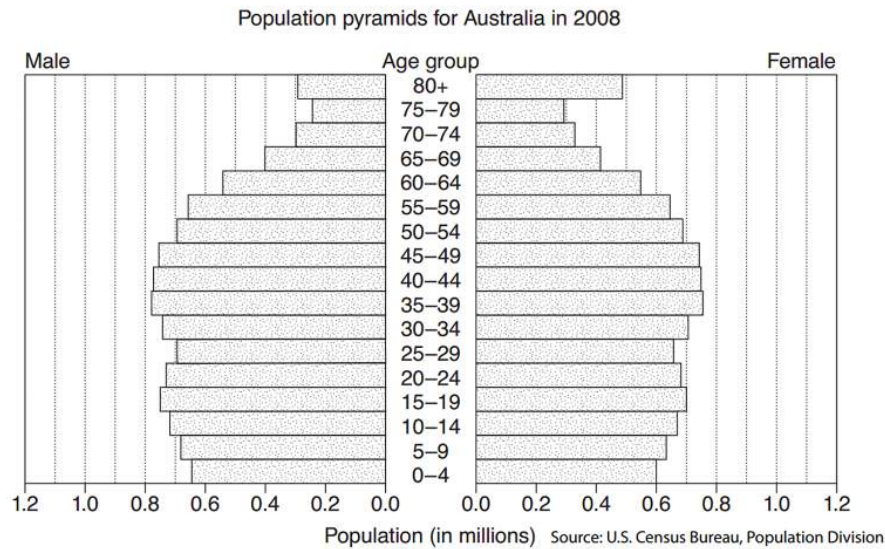
The graph displays the mean monthly rainfall in Sydney and Perth.



- i. For how many months is the mean monthly rainfall higher in Perth than in Sydney? (1 mark)
- ii. For which of the two cities is the standard deviation of the mean monthly rainfall smaller? Justify your answer WITHOUT calculations. (1 mark)

14. Statistics, STD2 S1 2008 HSC 26d

The graph shows the predicted population age distribution in Australia in 2008.

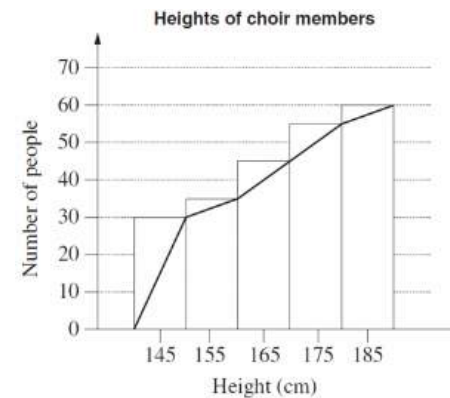


- How many females are in the 0-4 age group? (1 mark)
- What is the modal age group? (1 mark)
- How many people are in the 15-19 age group? (2 marks)
- Give ONE reason why there are more people in the 80+ age group than in the 75-79 age group. (1 mark)

15. Statistics, STD2 S1 2006 HSC 24c

The heights of the 60 members of a choir were recorded. These results were grouped and then displayed as a cumulative frequency histogram and polygon.

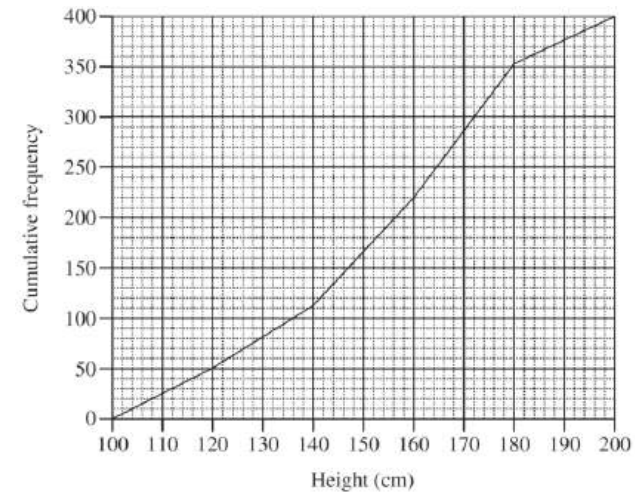
The shortest person in the choir is 140 cm and the tallest is 190 cm.



Draw an accurate box-and-whisker plot to represent the data. (3 marks)

16. Statistics, STD2 S1 2016 HSC 27c

The heights of 400 students were measured. The results are displayed in this cumulative frequency polygon.



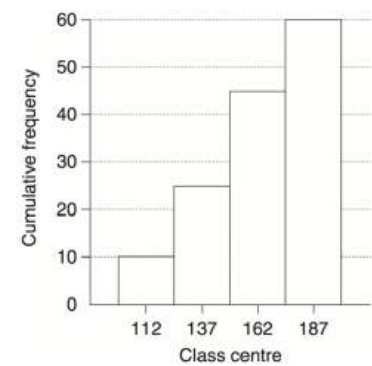
Use the polygon to estimate the interquartile range. (2 marks)

17. Statistics, STD2 S1 2010 HSC 26b

A new shopping centre has opened near a primary school. A survey is conducted to determine the number of motor vehicles that pass the school each afternoon between 2.30 pm and 4.00 pm.

The results for 60 days have been recorded in the table and are displayed in the cumulative frequency histogram.

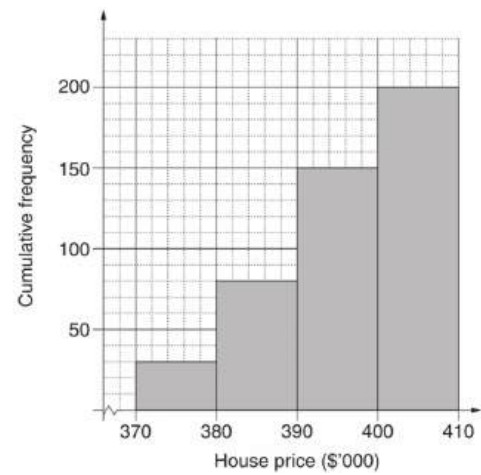
Score	Class centre	Frequency	Cumulative frequency
100–124	112	10	10
125–149	137	X	25
150–174	162	20	45
175–199	187	15	60



- i. Find the value of X in the table. (1 mark)
- ii. Carefully copy the cumulative frequency histogram into your writing booklet.
On the cumulative frequency histogram you have copied draw a cumulative frequency polygon (ogive) for this data. (1 mark)
- iii. Use your graph to determine the median. Show, by drawing lines on your graph, how you arrived at your answer. (1 mark)
- iv. Prior to the opening of the new shopping centre, the median number of motor vehicles passing the school between 2.30 pm and 4.00 pm was 57 vehicles per day.
What problem could arise from the change in the median number of motor vehicles passing the school before and after the opening of the new shopping centre?
Briefly recommend a solution to this problem. (2 marks)

18. Statistics, STD2 S1 2015 HSC 29d

Data from 200 recent house sales are grouped into class intervals and a cumulative frequency histogram is drawn.



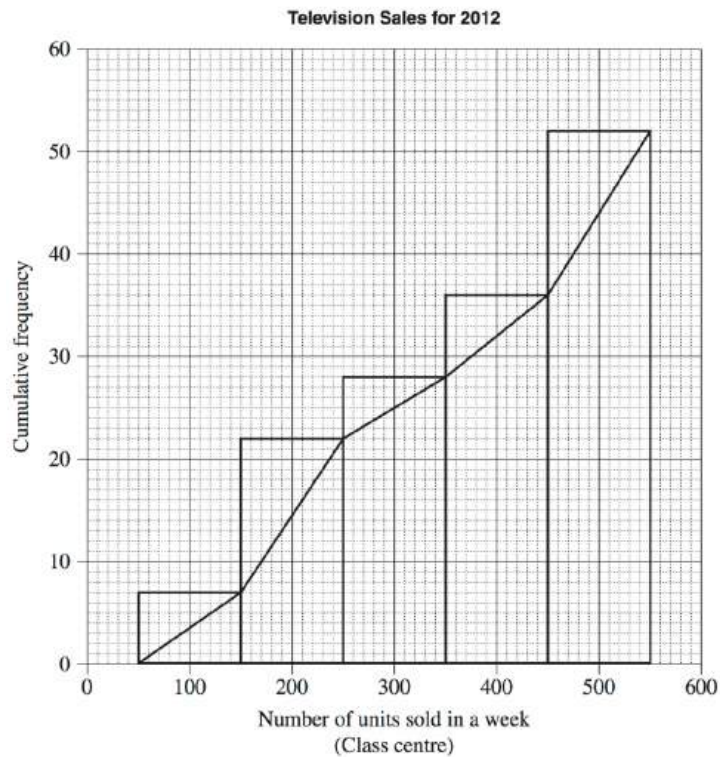
- i. Use the graph to estimate the median house price. (1 mark)
- ii. By completing the table, calculate the mean house price. (3 marks)

Class Centre (\$'000)	Frequency

19. Statistics, STD2 S1 2013 HSC 27c

A retailer has collected data on the number of televisions that he sold each week in 2012.

He grouped the data into classes and displayed the data using a cumulative frequency histogram and polygon (ogive).



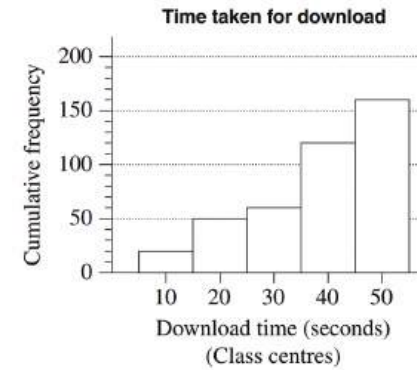
i. Use the cumulative frequency polygon to determine the interquartile range. **(2 marks)**

ii. Oscar said that the retailer sold 300 televisions in 6 of the weeks in 2012.

Is he correct? Give a reason for your answer. **(1 mark)**

20. Statistics, STD2 S1 2014 HSC 26e

The times taken for 160 music downloads were recorded, grouped into classes and then displayed using the cumulative frequency histogram shown.



On the diagram, draw the lines that are needed to find the median download time. **(2 marks)**

Worked Solutions

1. Statistics, 2ADV S2 2021 HSC 4 MC

The gradient of the cumulative frequency histogram will increase gradually, be steepest at day 10 then decrease gradually.

⇒ C

2. Statistics, STD2 S5 2010 HSC 4 MC

Normally distributed data have a frequency histogram graph that is shaped like a bell.

⇒ A

3. Statistics, STD2 S1 2011 HSC 11 MC

Mode of $X = 9$

Range of $X = 9 - 3 = 6$

Mode of $Y = 8$

Range of $Y = 11 - 5 = 6$

∴ X has a larger mode and ranges are the same

⇒ B

♦ Mean mark 47%

4. Statistics, STD2 S1 2013 HSC 15 MC

Total number of goals scored

$$= (3 \times 3) + (4 \times 7) + (5 \times 5) + (6 \times 1) + (7 \times 0) + (8 \times 4)$$

$$= 9 + 28 + 25 + 6 + 0 + 32$$

$$= 100$$

$$\text{Number of games} = 3 + 7 + 5 + 1 + 4 = 20$$

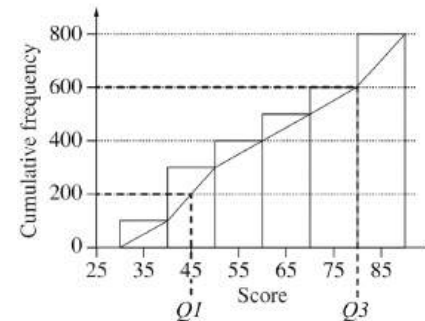
$$\therefore \text{Mean goals per game} = \frac{100}{20} = 5$$

⇒ C

5. Statistics, STD2 S1 2013 HSC 6 MC

D is the best as it starts at zero on the y -axis and has the same column widths.

6. Statistics, STD2 S1 2005 HSC 9 MC



$$IQR = Q3 - Q1$$

$$= 80 - 45$$

$$= 35$$

⇒ C

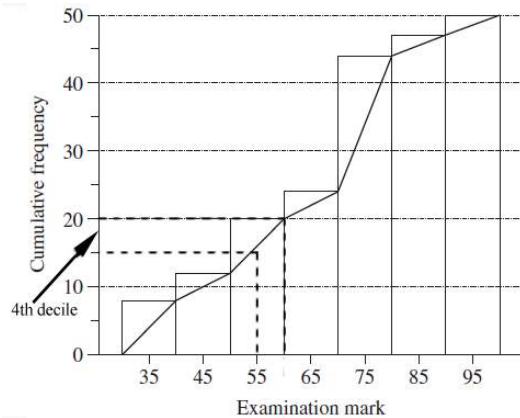
7. Statistics, STD2 S1 2020 HSC 7 MC

Positive skew occurs when the tail on the histogram is longer on the right-hand (positive) side.

♦♦ Mean mark 32%.

⇒ A

8. Statistics, STD2 S1 2007 HSC 22 MC



4th decile occurs when cumulative frequency is between 15 and 20.

∴ Examination mark must be between 55 and 60.

⇒ B

9. Probability, STD2 S2 2010 HSC 23c

i. # People = $5 + 15 + 10 + 3 + 1$
 $= 34$

ii. $P(B \text{ or } G) = P(B) + P(G)$
 $= \frac{5}{34} + \frac{10}{34}$
 $= \frac{15}{34}$

10. Statistics, STD2 S1 2011 HSC 25b

i. Year 12 (100%)

ii. % Ownership in Year 9 = $\frac{55}{70}$
 $= 78.6\% \text{ (1d.p.)}$

% Ownership in Year 10 = $\frac{50}{60}$
 $= 83.3\% \text{ (1d.p.)}$

∴ The Year 10 student is more likely to own a mobile phone.

iii. % Ownership increases as students progress from Year 7 to Year 12.

MARKER'S COMMENT: Many students ignore the instruction to **use calculations** and lose marks as a result.

11. Statistics, STD2 S1 2005 HSC 24d

i. Let the population of Sumcity = P

$15\% \times P = 24\,000$
 $\therefore P = \frac{24\,000}{0.15}$
 $= 160\,000 \dots \text{as required}$

ii. Western Suburbs population

$= 10\% \times 160\,000$
 $= 16\,000$

The column graph has this population as 12 000 people which is incorrect.

12. Statistics, STD2 S1 2006 HSC 24a

Reasons the graph is misleading include

- the columns are a different widths/volumes
- the vertical axis doesn't start at zero

13. Statistics, STD2 S1 2018 HSC 26d

- i. **3 months (Jul, Aug and Sep)**
- ii. **The mean monthly rainfall of Sydney is in a much tighter range than Perth.**
 $\therefore$ Sydney has a smaller standard deviation.

♦ Mean mark part (ii) 38%.
COMMENT: Extremely volatile result between parts with part (i) producing a 91% mean mark.

14. Statistics, STD2 S1 2008 HSC 26d

i. **# Females (0-4) = $0.6 \times 1\,000\,000$**
= 600 000

ii. **Modal age group = 35 – 39**

iii. **# Males (15-19) = $0.75 \times 1\,000\,000$**
= 750 000

Females (15-19) = $0.7 \times 1\,000\,000$
= 700 000

$\therefore$ Total People (15-19) = $750\,000 + 700\,000$
= 1 450 000

- iv. **The 80+ group includes all people over 80 and is not restricted by a 5-year limit.**

15. Statistics, STD2 S1 2006 HSC 24c

Low = 140

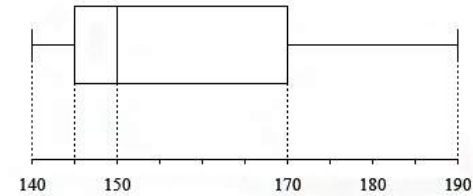
High = 190

Median = 150 (# People = 30)

$Q_1 = 145$ (# People = 15)

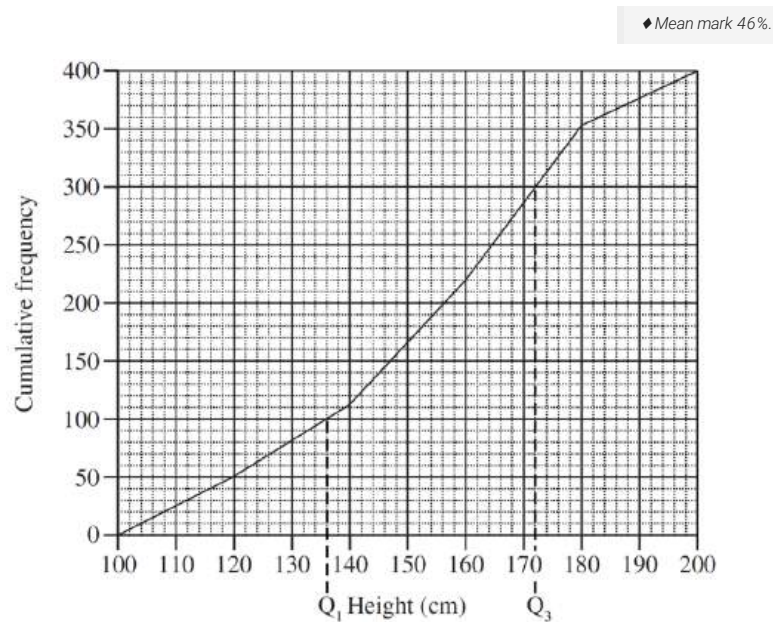
$Q_3 = 170$ (# People = 45)

Box and Whisker



16. Statistics, STD2 S1 2016 HSC 27c

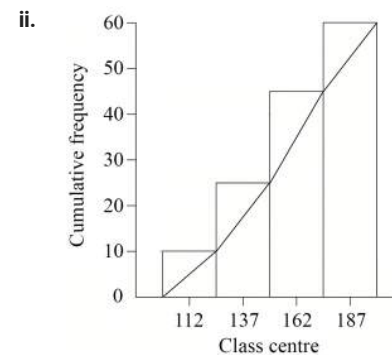
See graph for values:



$$\begin{aligned}
 IQR &= Q_3 - Q_1 \\
 &= 172 - 136 \\
 &= 36 \text{ cm}
 \end{aligned}$$

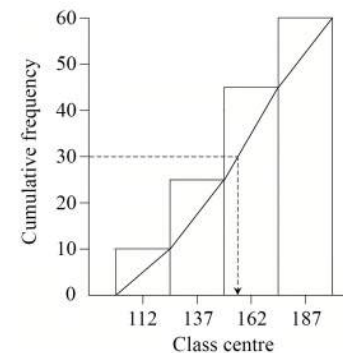
17. Statistics, STD2 S1 2010 HSC 26b

i. $X = 25 - 10$
 $= 15$



♦♦♦ Mean mark 18%
MARKER'S COMMENT: The ogive was poorly drawn with many students incorrectly joining the middle of each column rather than from corner to corner.

iii. Median ≈ 155



♦♦ Mean mark 25%
MARKER'S COMMENT: Many students did not "show by drawing lines on the graph" as the question asked.

- iv. **Problems**
- increased traffic delays
 - increased danger to students leaving school

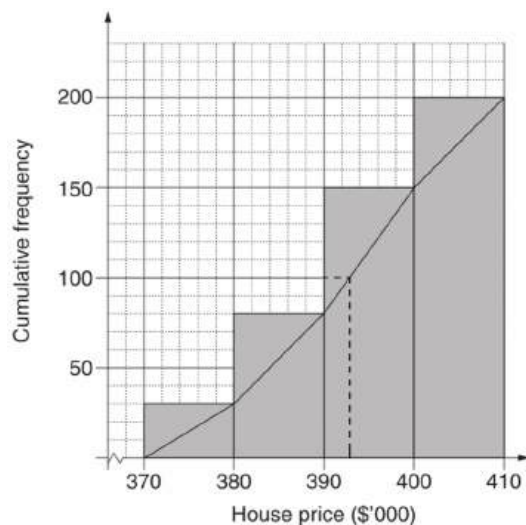
♦ Mean mark 47%
MARKER'S COMMENT: Short answers were often the best. Be concise when you can.

Solutions

- signpost alternative routes around school
- decrease the speed limit in the area

18. Statistics, STD2 S1 2015 HSC 29d

i.



From the graph, the estimated median house price = \$392 500

ii.

Class Centre (\$'000)	Frequency
375	30
385	50
395	70
405	50

♦♦♦ Mean marks of 9% for part (i) and 34% for part (ii)!

Mean house price (\$'000)

$$= \frac{375 \times 30 + 385 \times 50 + 395 \times 70 + 405 \times 50}{200}$$

$$= \$392$$

∴ Mean house price is \$392 000

19. Statistics, STD2 S1 2013 HSC 27c

i. Since cumulative frequency = 52 at max

$$\Rightarrow \text{Lower quartile at week} = \frac{52}{4} = 13$$

Lower quartile = 190 (from graph)

$$\Rightarrow \text{Upper quartile at week} = 3 \times 13 = 39$$

Upper quartile = 470

$$\therefore \text{IQR} = 470 - 190 \\ = 280$$

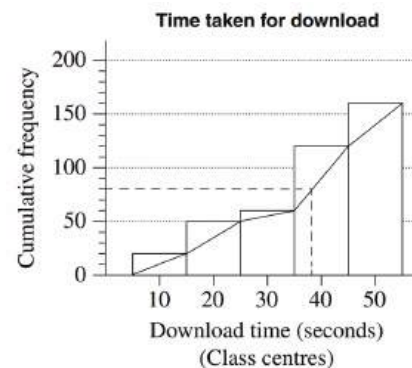
♦♦♦ Mean mark 13%
COMMENT: Finding median and quartile values from cumulative frequency charts is an important skill that is often examined.

ii. Oscar is not correct because the data is grouped.

He could only say the retailer sold between 250 and 350 units in 6 of the weeks in 2012.

♦♦♦ This question proved a beast, with a mean mark of 1%. Toughest question in the 2013 exam!

20. Statistics, STD2 S1 2014 HSC 26e



♦♦♦ Mean mark 17%.